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Extremely bending-resistant and compression-insensitive flexible circuits enabled by real-time crack healing via spontaneous interfacial wetting

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Abstract

Flexible circuits that allow stable electrical conductance and signal transmission under deformation are highly required for flexible electronics. However, bending-induced fatigue cracks in solid conductive materials can significantly reduce the stability of flexible circuits. Although liquid metal (LM) is a promising candidate for bending-resistant circuits, LM circuits are prone to being squeezed and susceptible to compression. Here, an extremely bending-resistant and compression-insensitive flexible circuit was achieved by the complementary effects of the fluidity of the upper LM and the good conductivity and rigidity of a thick bottom copper (Cu) layer. The fluidic LM can heal Cu cracks in real-time by spontaneous interfacial wetting to enhance the bending durability, whereas the solid Cu layer can maintain conductivity under compression to enhance the compression insensitivity. Compared with a conventional Cu circuit, such a flexible circuit achieves a 120-fold improvement in folding lifespan, while its resistance fluctuation under compression is only 0.05% of that of the counterpart LM circuit. Meanwhile, the bottom Cu pattern exactly facilitates the high-resolution selective printing of LM by reactive wetting. As a demonstration, a flexible signal acquisition system constructed with this circuit can continuously capture pulse signals during bending and remain functional after 10 000 bending cycles. The simultaneously enhanced bending stability and compression insensitivity greatly increase the applicability and reliability of flexible circuits under various deformation conditions.

Keywords: flexible circuits, extremely bending resistant, compression insensitive, real-time crack healing

1. Introduction

Flexible circuits are widely used in foldable optoelectronic displays [1-5], stretchable electronics [6-8], flexible sensors [9-12], and wearable medical diagnostic systems [13-15], serving for reliable electrical interconnection and stable electrical signal transmission. Owing to their high conductivity, mechanical deformability, and compatibility with high-resolution and high-throughput established manufacturing processes, laminated rigid metal thin films (copper (Cu), gold (Au), aluminum (Al), etc.) with polymer substrates (polyimide (PI), polyethylene terephthalate (PET), etc.) have long been mainstream configurations in current practical flexible circuits [16-19]. However, the inherent requirement for deformability of flexible electronics poses a challenge to the stability and durability of such flexible circuits, as repeated mechanical stress is prone to induce cracks within the metal films [20, 21]. Such undesirable mechanical failures can lead to resistance drift, fluctuations, or even complete loss in electrical conductivity, severely affecting the functions and lifespan of flexible electronics. To suppress crack propagation, great efforts have been made by inserting another interlayer [22-24] or enhancing interface adhesion [25-30], which can effectively reduce the practical strain on the metal film or prevent local stress concentration, enabling the flexible circuits to withstand greater applied strain without cracking. Even so, it is worth noting that repeated small strain cycles, even below the fracture strain of metal film, can also induce fatigue cracks during long-term stress cycles due to the mechanical mismatch between the stiff metal film and the compliant substrate [20, 31]. Therefore, an effective strategy to improve long-term fatigue durability and prolong the lifespan of flexible circuits is urgently needed.

In recent years, due to its high fluidity and self-healing capability, liquid metal (LM) has emerged as a promising candidate for enhancing the durability of flexible circuits [32]. A common configuration is LM-elastomer composite featured LM microcapsules embedded in the elastomer matrix, which can release LM and bridge the conductive pathways to restore electrical functionality when the LM microcapsules are damaged [33-36]. However, patterning such materials into conductive flexible circuits still faces limitations in resolution and efficiency, such as the limited resolution of stencil printing, complex matching of multiple process parameters (nozzle diameter, nozzle height, flow speed, etc.), and inherent efficiency constraints of serial inkjet printing [37-41]. In recent years, laser scribing has also been employed as an effective method for circuit patterning due to its mask-free advantage [42-45]. However, this process may face issues such as droplet spreading or the risk of ablating the substrate. Reactive wetting with metal films offers an efficient method for printing high-resolution pure LM circuits [46-50]. By inheriting the inherent high precision of the photolithography process [51, 52], this method can be used to fabricate very fine LM circuits based on high-precision metal patterns prepared by photolithography. Such LM circuits offer excellent stretching and bending performances, but an often neglected issue is their susceptibility to compression. The conductivity of such circuits is provided mainly by the thicker LM layer, as the metal film is often very thin (at the

nanoscale) and may undergo an alloying reaction with LM. However, owing to the high fluidity, LM conductive path is prone to being squeezed, making the resistance of the circuit highly susceptible to compression, which hinders its broad adaptability under various deformation conditions.

Here, we propose an extremely bending-resistant and compression-insensitive flexible circuit by the complementary effects of the fluidity of the upper LM and the good conductivity and rigidity of a thick bottom copper (Cu) layer. The highly fluidic LM can timely heal the cracks in the Cu layer during bending by spontaneous interfacial wetting to enhance the bending durability, whereas the highly conductive thick Cu layer can maintain conductivity under compression to enhance the compression insensitivity. Compared with the counterpart Cu circuits, the obtained flexible circuits exhibit a durability of 230 000 cycles under a bending radius of 0.5 mm and 60 000 cycles under repeated folding conditions, achieving an improvement of 2.3 and 120 times, respectively. Moreover, its resistance fluctuation under compression (4 MPa) is only 0.05% that of the counterpart LM circuit. Meanwhile, the bottom Cu pattern exactly facilitates the high-resolution and high-efficiency selective printing of LM by reactive wetting. As a proof of concept, such a robust flexible circuit was integrated into a flexible system for acquiring human pulse signals, which can continuously collect pulse signals during bending with negligible interference and operate normally even after 10 000 bending cycles. The excellent bending durability and compression insensitivity greatly enhance the applicability and reliability of flexible circuits under various deformation conditions.

2. Results and discussion

Figure 1(a) illustrates the structure and advantage of the bending-resistant and compression-insensitive flexible circuit enabled by real-time crack healing of the bottom Cu layer by spontaneous interfacial wetting of the upper LM. For conventional flexible circuits consisting of rigid metal films and polymer substrates, cracks often propagate in the metal films during repeated bending cycles due to interface delamination, excessive local stress, or fatigue damage (Figure 1(a-i)), leading to abrupt resistance change or even complete loss of electrical functionality, which severely affects the continuous operation stability and lifespan of the electronic systems. As a potential alternative to solid metal circuits, LM circuits exhibit good bending stability owing to their excellent fluidity. However, this also causes the LM channels to be prone to undergoing a significant reduction in cross-sectional area under compression, making the resistance of the circuit highly susceptible to pressure (Figure 1(a-ii)). To address this conflict, we propose a bending-resistant and compression-insensitive flexible circuit consisting of a bottom thick Cu layer and an LM healing layer (Figure 1(a-iii)). When cracks propagate in the Cu film during bending, LM can fill and bridge the cracks in real-time through spontaneous interfacial wetting, thus enhancing the bending stability of the flexible circuits. Meanwhile, the rigid thick Cu layer can withstand higher pressure under compression, which enables the resistance of the circuit being less affected by squeezing, thereby achieving a win-win combination of excellent bending durability and compression insensitivity. A comparison of the scanning electron microscope (SEM) images of the Cu circuit

Template for the IJEM
and our Cu-LM circuit after bending confirms the role of LM in healing fatigue cracks (Figure 1(b)). Figure 1(c) shows an image of the Cu-LM flexible circuit after being stored for 60 days in an ambient environment. The upper surface and bottom layer of the circuit still exhibited distinguishable colors of LM and Cu, proving that the bilayer metal films can maintain their respective morphologies stably over a long time without undergoing serious alloying reactions (details are discussed in Figure 3). Moreover, the bending-resistant flexible circuits are compatible with different commercial standard connectors to achieve stable electrical interconnection (Figure S1). Taking advantage of the excellent electrical stability, this flexible circuit can interconnect various electronic components to integrate into a flexible signal acquisition system for human pulse (Figure 1(d)), which can stably acquire pulse signals of humans even after 10 000 bending cycles. Figure 1(e) shows the comparison of the cyclic folding durability between the Cu-LM flexible circuit and the conventional Cu circuit. The results show that the resistance of the conventional Cu circuit began to fluctuate dramatically after a few hundred cycles, whereas the Cu-LM circuit maintained extremely high stability over a long period of time. Specifically, the relative resistance change (in the folded state) of the conventional Cu circuit reached more than 300% after just 500 folding cycles, while that of the Cu-LM circuit could remain below 300% even after 60 000 folding cycles, thereby achieving at least a 120-fold improvement in the folding lifespan. These results indicate the significant role of real-time-healing cracks in enhancing stability and prolonging the lifespan of flexible circuits. Figure 1(f) shows a comparison of the resistance changes between the Cu-LM circuit and the counterpart LM circuit (with only 50 nm of Cu for the selective printing of LM) under compression. Under a pressure of 4 MPa, the resistance of the LM circuit increased by 208.4%, whereas that of the Cu-LM circuit increased by only 0.11%, demonstrating that the thick Cu layer plays an important role in enhancing the compression insensitivity of the circuit. As shown in Figure 1(g), the bending durability of our flexible circuit is superior to the reported performances of other flexible circuits [16, 17, 21, 30, 53-59], as it can withstand more bending cycles under extremely small bending radii (<1 mm) while maintaining very low resistance changes. Such excellent bending durability and conductivity are attributed to the real-time healing of cracks by the highly fluidic LM, as well as the inherently excellent electrical conductivity and ability to maintain the shape of solid Cu film under extremely small bending radii, which avoids possible flow interruption of pure LM due to squeezing of the channel under extremely small bending radii [56].

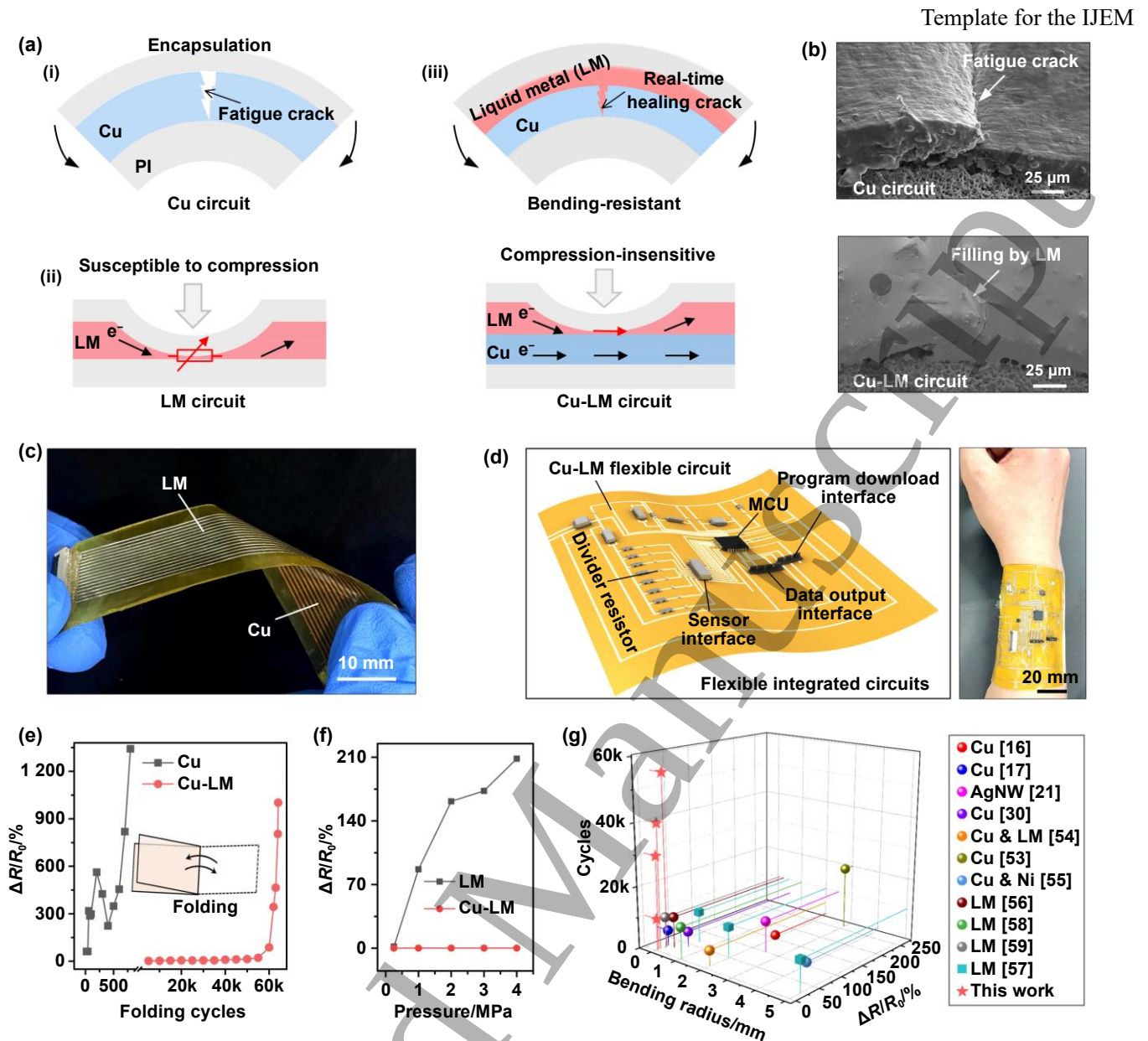


Figure 1. Mechanism and structure of the real-time healing bending-resistant and compression-insensitive flexible circuit. (a) Mechanism and advantage of the bending-resistant and compression-insensitive flexible circuit (iii) compared to conventional Cu circuit (i) and LM circuit (ii). (b) SEM images of the conventional Cu circuit and our Cu-LM circuit after cyclic bending. (c) Optical image of the flexible circuit. (d) Application demonstration of the flexible circuit for wearable signal acquisition system for human pulse by interconnecting with multiple electronic components. (e) Enhanced stability and extended lifespan of the Cu-LM flexible circuit compared to conventional Cu circuit. (f) Enhanced compression insensitivity of the Cu-LM flexible circuit compared to conventional LM circuit. (g) Comparison of the bending durability with other reported flexible circuits.

High-resolution and high-efficiency manufacturing is a prerequisite for the widespread applications of the flexible circuits. Figure 2 shows the principle and capability of the simple and high-resolution method for printing LM, enabled by selective wetting on Cu patterns through simple dip-coating in a naturally stratified

solution of sodium hydroxide (NaOH) and bulk LM. Here, the Cu pattern was obtained by photolithography and wet etching (Figures S2 and S3). The Cu/PI film we used is the commercially available flexible copper clad laminate (FCCL), i.e., PI-Cu laminate, which is used for manufacturing flexible PCBs in the industry. This Cu/PI film exhibits excellent adhesion between Cu and PI. We tested the adhesion between Cu and PI by the Scotch tape peeling test [60] and pull-off test [61]. As shown in Movie S1, the etched Cu pattern remains intact after the Scotch tape peeling test. Moreover, in the pull-off test (Figure S4 and Movie S2), the Cu layer did not detach from the PI substrate even under a maximum pulling force of 14.429 MPa. No residual Cu was observed on the dolly, and the failure occurred preferentially at the interface between the PI and the underlying steel block. Such strong adhesion between Cu and PI is beneficial for the bending performance of the flexible circuit. The sticky surface oxides of LM often adhere to various surfaces without selectivity, making it difficult to manipulate and pattern LM [62]. Considering that oxides can react with acid or alkaline solutions [63, 64], we dripped LM into a solution of NaOH. Due to the higher density of LM, it sank beneath the NaOH solution, forming a distinctly two-layer solution. Fortunately, during this natural stratification process, the surface oxides of LM were removed by the NaOH solution. In this way, by a simple and efficient dip-coating method, the non-oxidized LM can selectively adhere onto the surface of the Cu pattern without leaving residue on the PI surface (Figure 2(a-i)). The significant difference in adhesion between Cu and PI surfaces is due to the distinct wettability of the non-oxidized LM on these two surfaces. As shown in Figure 2(a-ii), the LM droplet treated with NaOH (aq) exhibits non-wetting behavior on the PI surface with a contact angle of 127° , whereas it can wet well on the Cu film with a contact angle of 25° . This wettability difference enables the selective adhesion of the LM to the Cu surface, as seen from the comparison of optical images before and after dip-coating in Figure 2(a-iii). LM completely covered the Cu surface with clear edges, while there was no residue on the PI substrate. Therefore, the selective wetting behavior of LM on the Cu pattern facilitates the printing of LM, allowing it to serve as a downstream process compatible with traditional high-resolution and high-efficiency photolithography and etching processes.

The uniformity of LM adhering to the Cu surface is critical to the consistent electrical properties of flexible circuits. We observed that the spontaneously wetted LM on the Cu surface formed a relatively regular arc shape, which remained stable due to the formation of surface oxides upon exposure to air. Therefore, the height of the LM exhibits an approximate geometric relationship determined by the line width and contact angle, as shown in Figure 2(b). To verify this relationship, a series of lines with different widths (Figure 2(c)) were printed, and the heights of the LM were measured by laser confocal microscope. The experimental results are generally in agreement with the theoretical curve, indicating the correctness of the geometric relationship (Figure 2(d)). This self-wetting-dependent geometric relationship provides a foundation for the uniform amount of LM printed on the Cu surface and enhances the controllability of printing. The applicability of this method for printing patterns across a wide span of line widths was further demonstrated by

various patterns, ranging from microscopic high-resolution lines, dots, and circles (Figure 2(e)) to macroscopic patterns (Figure S5). This method can achieve a minimum line width of approximately 35 μm and a minimum line space of 35 μm (Figure S6). Therefore, the Cu pattern template effectively simplifies the printing of LM with different line widths by selective interfacial wetting, avoiding the complexity of line width control and limited resolution in LM printing, while the upper LM in the obtained bilayer circuit can enhance the bending stability of the flexible circuit by real-time healing of cracks, achieving a win-win situation of convenient manufacturing and excellent performance.

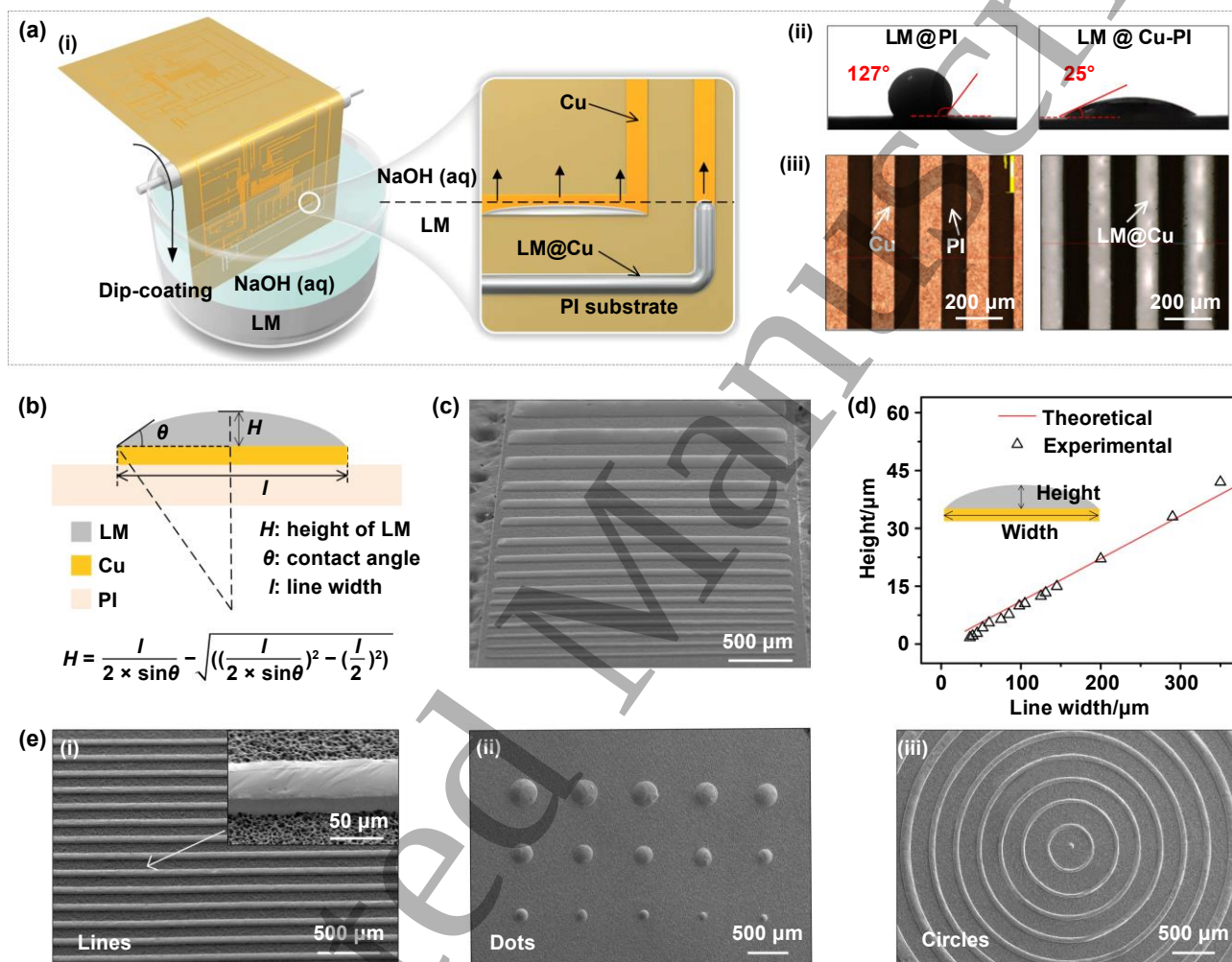


Figure 2. Patterning LM by a simple dip-coating method via selective interfacial wetting. (a) Patterning principle of Cu-LM circuits. (i) Dip-coating LM on the Cu pattern in a naturally stratified solution of NaOH (top layer) and bulk LM (bottom layer) by selective wetting. (ii) Contact angles of LM droplets on PI substrate (left) and Cu film (right). (iii) Optical images of Cu pattern on PI substrate (left) and the corresponding Cu-LM pattern after dip-coating LM (right). (b) The height of LM layer determined by the line width and contact angle. (c) SEM images of Cu-LM lines with a series of different widths. (d) The theoretical and experimental relationships between the line width and height of LM. (e) SEM images of different high-resolution Cu-LM patterns.

The capability to fabricate large-area complex patterns with various line widths is of great significance for the efficient manufacturing of complex flexible circuits. Meanwhile, whether LM and Cu can coexist stably over the long term is also an issue that needs to be explored, as it relates to whether they can maintain their respective liquid and solid states to fulfill their complementary roles in enhancing bending stability and compression insensitivity. As a demonstration, we fabricated a large-area complex pattern of traditional Chinese gardens through a simple dip-coating process. As shown in Figure 3(a), LM fully covered all parts of the Cu patterns with different line widths, demonstrating the feasibility and high efficiency of this method for printing large-area complex patterns. To investigate whether LM and Cu can coexist stably over a long period of time, we analyzed the morphology and interfacial element composition of the flexible circuit stored for 60 days in an ambient environment. As shown in Figure 3(a-iii), the LM and Cu visually retained their respective original colors. Furthermore, the element compositions of three different locations of the cross-section of the Cu-LM flexible circuit were analyzed by field emission scanning electron microscope (SEM) and energy dispersive spectroscopy (EDS). Figure 3(b-i) shows the element composition of the top layer. Three elements, gallium (Ga), indium (In), and tin (Sn), were present in this layer, which corresponded exactly to the three components of the LM. At the interface between Cu and LM, both Cu and Ga were present, confirming the occurrence of an alloying reaction between LM and Cu [65]. However, in the bottom layer, only Cu was observed, indicating that the alloying reaction did not penetrate further downward. The element composition at the above three locations indicates that the alloying reaction occurred only at the interface between the LM and Cu after long-term storage, while the top layer and the bottom layer still maintained their original bulk LM and Cu, respectively. This indicates that the two layers of LM and Cu in the flexible circuit can coexist for a long time. This is attributed to the fact that the reaction between LM and Cu stops after their reaction product (CuGa_2) reaches a certain thickness, as CuGa_2 acts as a barrier to further reaction [50, 66-69]. The thickness of the CuGa_2 layer is typically 3–5 μm at room temperature [50, 69]. Consequently, conventional nanoscale metal layers may be fully dissolved by LM, which mainly acts as a wetting layer for printing LM. In contrast, our thick Cu layer (13 μm) retains a portion of the unreacted Cu at the bottom, which can continue to provide electrical conductivity. It is worth noting that the Cu-Ga compounds formed at the interface can effectively drive the wetting of LM on the Cu surface [50]. Therefore, this interfacial alloying reaction is actually beneficial, as it can enhance the wettability and adhesion between LM and Cu, which is helpful for the real-time healing of cracks generated on the Cu film and improving the bending durability of the flexible circuit.

To further investigate the long-term stability of the Cu-LM circuit, we also monitored the electrical stability of an encapsulated Cu-LM flexible circuit over three months by measuring its resistance every 10 days. The test results show that the resistance of the circuit did not obviously change during the three months (Figure S7), demonstrating that the flexible circuit can remain stable over a long time. This stable resistance

of the Cu-LM circuit is attributed to the ability of Cu and LM to coexist for a long time. In addition, the encapsulation also helps to isolate air and water vapor, preventing LM from being further affected by air and humidity.

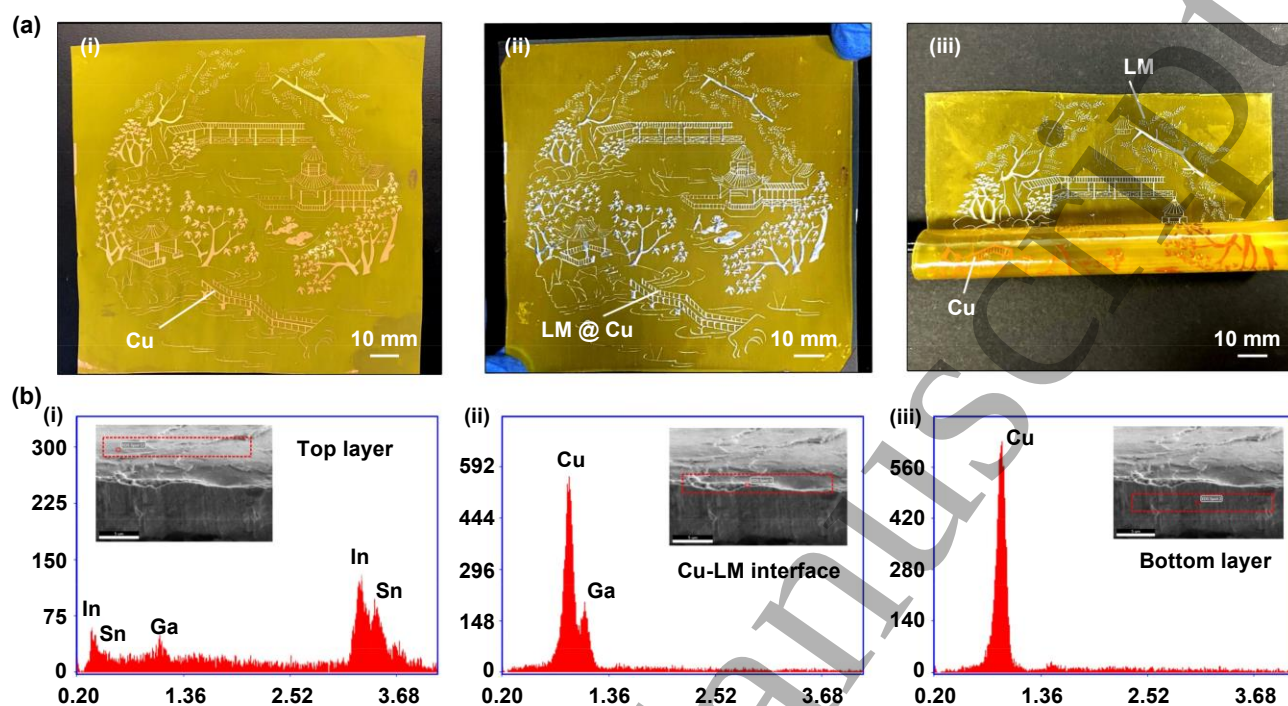


Figure 3. Large-area manufacturing and long-term stability of the Cu-LM pattern. (a) Optical images of a complex pattern of traditional Chinese gardens. (i) Wet-etched Cu pattern. (ii) Corresponding Cu-LM pattern. (iii) The front and back sides of the Cu-LM pattern. (b) Elemental analysis of the top layer (i), the interface between LM and Cu (ii), and the bottom layer (iii).

Benefiting from the real-time healing of cracks, compared with that of the conventional Cu circuit, the bending durability of the Cu-LM flexible circuit is significantly improved. To investigate the bending durability of the circuits, we tested the cyclic bending performance of these two types of flexible circuits at a bending radius of 0.5 mm (Figure 4(a)) and a bending frequency of 5 Hz. The initial resistance of the Cu circuit is approximately 2 Ω , and that of the Cu-LM circuit is approximately 1.4 Ω . As shown in Figure 4(b), during the bending process, the resistance changes of both circuits generally went through three stages, namely, the stable stage, the stage of dramatic increase in resistance, and the stage of failure. For the Cu circuit, the resistance began to increase rapidly after approximately 99 000 cycles, with a maximum resistance change of 912% (Figure 4(b-i)). In contrast, the stable stage of the Cu-LM circuit can persist up to approximately 230 000 cycles, with a maximum resistance change of only 130% (Figure 4(b-ii)). By comparing the resistance changes of these two circuits during the bending fatigue tests (Figure 4(c)), we find that the bending durability of the Cu-LM circuit is significantly better than that of its counterpart Cu circuit. This improvement in bending resistance is attributed to the real-time healing of fatigue cracks by LM. As shown in Figure 4(d), by observing the unencapsulated circuits subjected to the same bending cycles as the

encapsulated circuits in the above bending electromechanical tests, we found that obvious cracks propagated at the Cu circuit after cyclic bending (Figure 4(d-i)). Such cracks will open up under bending and further propagate with increasing bending cycles, leading to a sharp increase in resistance and eventual electrical failure. However, although cracks also appeared in the bottom Cu layer of the Cu-LM circuit, they were healed by LM via spontaneous interfacial wetting (Figure 4(d-ii)).

To elucidate the healing mechanism of the circuit, we observed the dynamic filling process of LM at the cracks in the Cu film under an optical microscope. Here, a thicker Cu film (300 μm) was applied for observation because it can generate more noticeable cracks under fewer bending cycles, thus facilitating the capture of the dynamic cracking and filling process in a shorter time. As shown in Movie S3 and Figure 4(e), as the number of bending cycles increased, cracks gradually initiated and expanded in the Cu film. Meanwhile, LM spontaneously filled them in real-time. Moreover, LM deformed conformally during the dynamic opening and closing of the crack, and no voids were observed at the interface between LM and the crack, indicating the good wettability of LM at the Cu crack interface. Therefore, LM can fill the Cu cracks and heal the conductive pathways, thereby extending the bending lifespan of the circuit. The increase in resistance before complete fracture, as shown in Figure 4(b-ii), may be attributed to the electrical conductivity of LM being lower than that of Cu. Therefore, after the crack is filled with LM, the resistance slightly increases compared with that of the original continuous and intact Cu layer. We further analyzed the possible reasons for the ultimate failure of the circuit (Figure S8). As the number of bending cycles further increases, the cracks in the Cu layer further propagate and grow in size, and slight interface delamination may occur in the Cu layer near the crack due to the high stress around the crack (as shown in Figure 1(b)). This can cause the Cu crack to open significantly during bending so that LM is unable to fully fill such a large crack gap in the bending state, leading to a large change in resistance and ultimate failure of the circuit.

We further investigated the effect of the bending frequency on the stability of the circuit. As shown in Figure 4(f)–(h), with increasing bending frequency, the bending durability of the circuit decreases. Specifically, at a bending frequency of 20 Hz, the resistance increased by 396% after 193 000 cycles. When the frequency was increased to 40 Hz, the relative resistance change reached 550% after 164 000 cycles. We attribute the decrease in the durability of the circuit with increasing bending frequency to two possible reasons. On the one hand, a higher bending frequency may influence immediate repair due to the viscoelasticity and inertial effects of LM. On the other hand, an increased bending frequency may accelerate the propagation of cracks in the Cu film, thereby reducing the bending durability of the circuit.

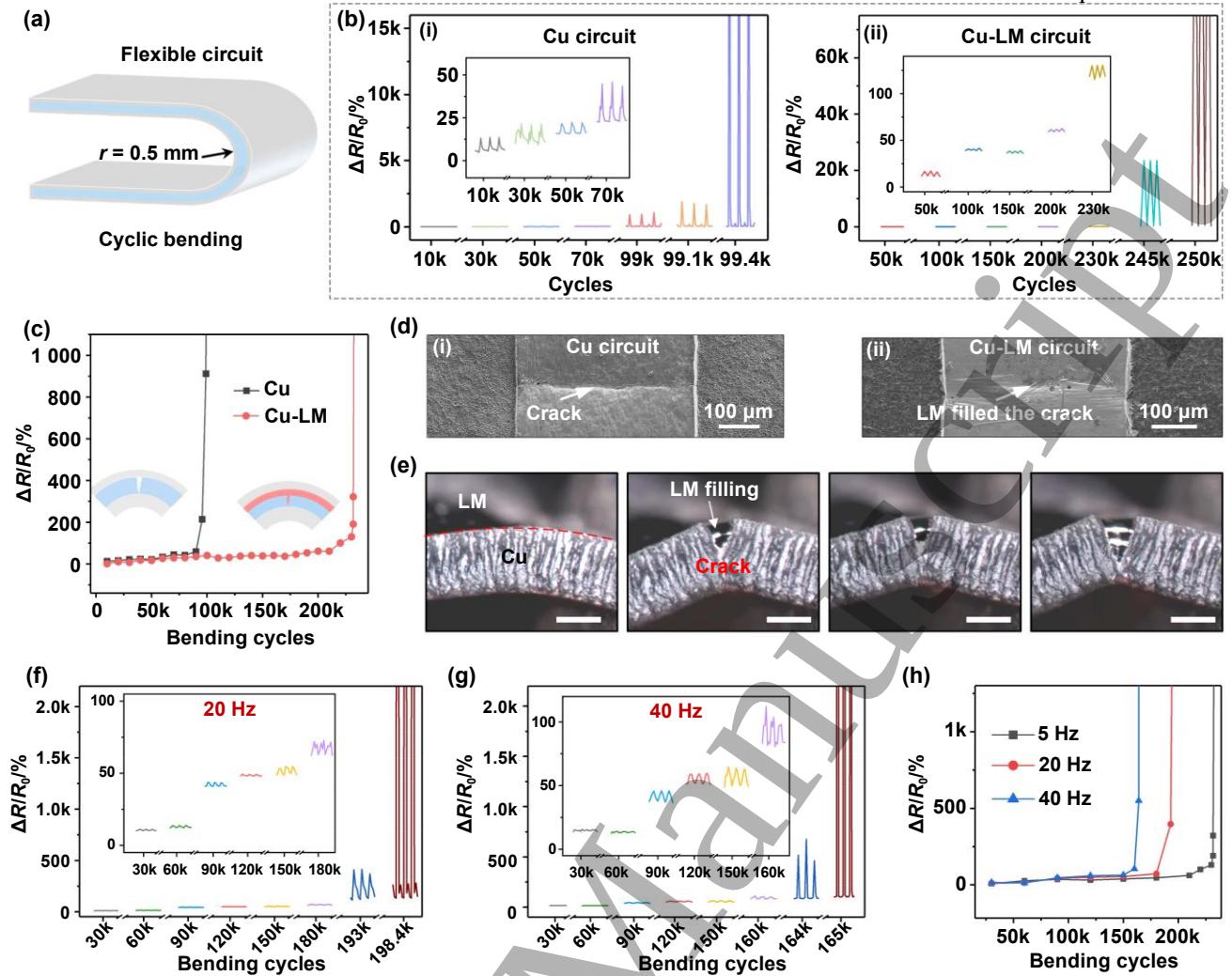


Figure 4. Bending performances of the Cu-LM flexible circuits. (a) Schematic illustration of the cyclic bending test at a radius of 0.5 mm. (b) Bending tests of Cu circuit (i) and Cu-LM circuit (ii). (c) Comparison of the long-term bending performances of Cu circuit and Cu-LM circuit. (d) SEM images of the unencapsulated circuits after bending. (i) Cu circuit. (ii) Cu-LM circuit. (e) Dynamic filling of cracks by LM. Scale bars, 200 μ m. (f) Bending durability of the Cu-LM circuit at a bending frequency of 20 Hz. (g) Bending durability of the Cu-LM circuit at a bending frequency of 40 Hz. (h) Influence of the bending frequency on the durability of the circuit.

To investigate the performance of the Cu-LM flexible circuit under harsher conditions, we further tested the long-term cyclic folding performance (Figure 5(a)). Here, the PI substrate was positioned on the inner side, while the polydimethylsiloxane (PDMS) encapsulation was on the outer side in the folded state, which is denoted as outer folding. As shown in Figure 5(b-i), the resistance variation of Cu circuit under folding far exceeded that during bending at a radius of 0.5 mm, indicating that folding is more likely to cause damage to the flexible circuit, with the resistance change rate surpassing 300% after only 500 cycles. The SEM image of the Cu circuit (without encapsulation) after the same folding cycles shown in Figure 5(b-ii) also confirms this point, as we can see that folding induced a wider crack than bending at a radius of 0.5 mm in Figure

4(d-i). However, the resistance variation of the Cu-LM circuit could remain below 300% even after 60 000 folding cycles (Figure 5(b-iii)), achieving at least a 120-fold improvement in the folding lifespan compared with the Cu circuit (Figure 5(c)). This enhancement of stability and lifespan under harsh conditions is attributed to the healing of cracks by LM (Figure 5(b-iv)). The healing of cracks enables the Cu-LM circuit to maintain the conductive pathways over a longer time, showing great significance for operational stability under deformations in practical applications.

The direction of folding is also an important factor affecting the folding lifespan of flexible circuits. To investigate the effect of folding direction on lifespan, we conducted a folding test in the opposite direction, i.e., “inner folding”. As shown in Figure S9, the flexible circuit exhibits a longer folding lifespan under inner folding compared to outer folding. After 60 000 cycles of outer folding, the relative resistance change reaches 87.4%. In comparison, the circuit can withstand 90 000 cycles under inner folding before reaching a similar degree of degradation (89.7%). The shorter lifespan under outer folding may be attributed to the easier initiation of cracks during outer folding.

In addition to the enhanced bending durability, the Cu-LM flexible circuit also exhibits excellent pressure insensitivity compared with the conventional pure LM circuit, as the solid Cu layer can maintain conductive pathways under pressure. To investigate the influence of compression on the circuits, we applied varying pressures to the circuit using a 3 mm × 3 mm indenter by a uniaxial mechanical testing machine (Figure 5(d)). As shown in Figure 5(e) and (f), for conventional LM circuits (with only a very thin bottom Cu layer (50 nm) for selective printing LM), the resistance clearly increased with increasing pressure. In contrast, the resistance of the Cu-LM circuit changed only slightly in response to increased pressure. Specifically, under a pressure of 4 MPa, the relative resistance change of the conventional LM circuit reached 208.4%. In contrast, the resistance of the Cu-LM circuit increased by 0.11%, which is only 0.05% of that of the LM circuit. Movie S4 also intuitively shows the excellent compression insensitivity of the Cu-LM circuit compared with the LM circuit during finger pressing. We attributed the compression insensitivity of the Cu-LM circuit to the fact that compared with the LM, the thick Cu layer can withstand much higher pressure under compression, as Cu has a high yield strength on the order of tens to hundreds of MPa. For conventional LM circuits, the local pressure is prone to reducing the cross-sectional area of the LM channel due to the fluidity of LM, thereby leading to a large resistance increase of the circuit. It should be noted that although solid metal layers are also present beneath the LM in conventional LM circuits, they are typically only nanometer-scale in thickness and mainly used for selectively printing LM, whereas the conductivity of the circuit is determined mainly by the thicker LM [46, 47, 49]. In contrast, for our Cu-LM circuit, the solid Cu layer (with a thickness of 13 μm) with good conductivity and the LM layer simultaneously provide conductivity. Therefore, although the cross-sectional area of the LM layer decreases under pressure, the Cu layer can still maintain the conductive pathways, thus preventing significant fluctuations in the resistance of

the circuit. Therefore, the thick solid Cu layer at the bottom plays a unique role in enhancing the stability of the circuit under compression.

To conduct a more comprehensive assessment of the performance of the flexible circuit, we further tested its stability under various deformation conditions and environmental conditions. The Cu-LM circuit exhibits good stability under other deformation conditions, such as stretching and twisting. As shown in Figure S10, with the increase of strain, the resistance of the circuit increased slightly. Under 5% strain, the relative resistance change was approximately 0.99%, demonstrating good stretching stability. Moreover, we tested the resistance of the circuit at different twisting angles. As shown in Figure S11, the testing results indicate that as the twisting angle increases, the resistance shows negligible changes, demonstrating that the circuit has good stability under twisting conditions. For practical applications, it is also necessary to test the impact of temperature and humidity on the circuit. To investigate the effect of temperature on the stability of the circuit, we tested its resistance from ambient temperature (approximately 23 °C) to 200 °C. As shown in Figure S12, the resistance of the circuit increased slightly with increasing temperature. The increase in resistance is mainly due to the fact that the resistivity of both Cu and LM increase with temperature. In addition, it should be noted that too low temperature may affect the healing efficiency of LM due to the increase in viscosity and the reduced rate of the alloying reaction between LM and Cu [70]. Therefore, Cu-LM circuit is more suitable for long-term use at moderate temperatures. As reported by previous studies [71, 72], high humidity can accelerate the oxidation of LM and thicken the oxide layer. However, for our flexible circuit, the Cu-LM conductive layer is encapsulated with PDMS, which can theoretically isolate the air and water vapor to prevent the circuit from being affected by humidity. To validate whether high humidity affects our encapsulated circuit, we immersed the circuit in water for 10 days. As shown in Figure S13, the resistance of the circuit changed negligibly over 10 days, demonstrating that the encapsulation is reliable and can prevent the circuit from being affected by humidity.

In practical applications, despite the low toxicity of gallium-based LM, its leakage may lead to other issues, such as short circuits or electrical failures, affecting the stability of the circuits. Hence, we also investigated the leakage resistance of our flexible circuit. As shown in Figure S14, we applied increasing pressures to the circuit using a uniaxial mechanical testing machine and observed the morphology of the circuit after each pressure loading-unloading process. The testing results indicate that the leakage resistance of the circuit can reach approximately 4 MPa, which is comparable to other reported LM circuits [59, 73-76] (Table S1). This excellent leakage resistance is related to the good encapsulation between PI and PDMS. As shown in Figure S15, we tested the adhesion between PI and PDMS by peel test [77, 78]. Here, a 10 mm wide PI-PDMS sample was used for the peel test, and the moving speed of the clamp was 50 mm·min⁻¹. The peel force remained relatively stable at approximately 2.7 N. Therefore, the peel strength between PI and PDMS is approximately 0.27 N·mm⁻¹, which is approximately 26 times greater than that of PI-PDMS

adhesion by ultraviolet (UV) ozone surface treatment [77]. This high adhesion can be attributed to the porous microstructure on the surface of the PI film in the PI/Cu laminate we used. As shown in Figure S15(d), stretched PDMS micro-pillar structures can be observed at the peel interface between PI and PDMS, reflecting the embedding of PDMS into the porous structure of PI, which can enhance interfacial adhesion by mechanical anchoring. The reliable encapsulation is also related to the fact that PDMS can effectively cover the LM circuit. This is attributed to the wetting of LM oxide layer on PDMS. As shown in Figure S16, the contact angle of the LM droplet (with oxide skin) on PDMS is slightly less than 90° . The oxide layer of LM enables the LM droplets to adhere to the PDMS surface. In comparison, the contact angle of pure LM (after treatment with HCl) on PDMS is approximately 152° , indicating that pure LM exhibits poor wettability on the PDMS surface. For our Cu-LM circuits, before the PDMS encapsulation process, LM was exposed to air for a period of time, which resulted in the formation of a surface oxide layer. Due to the good adhesion between the LM oxide layer and PDMS, PDMS can effectively encapsulate the LM circuits [48].

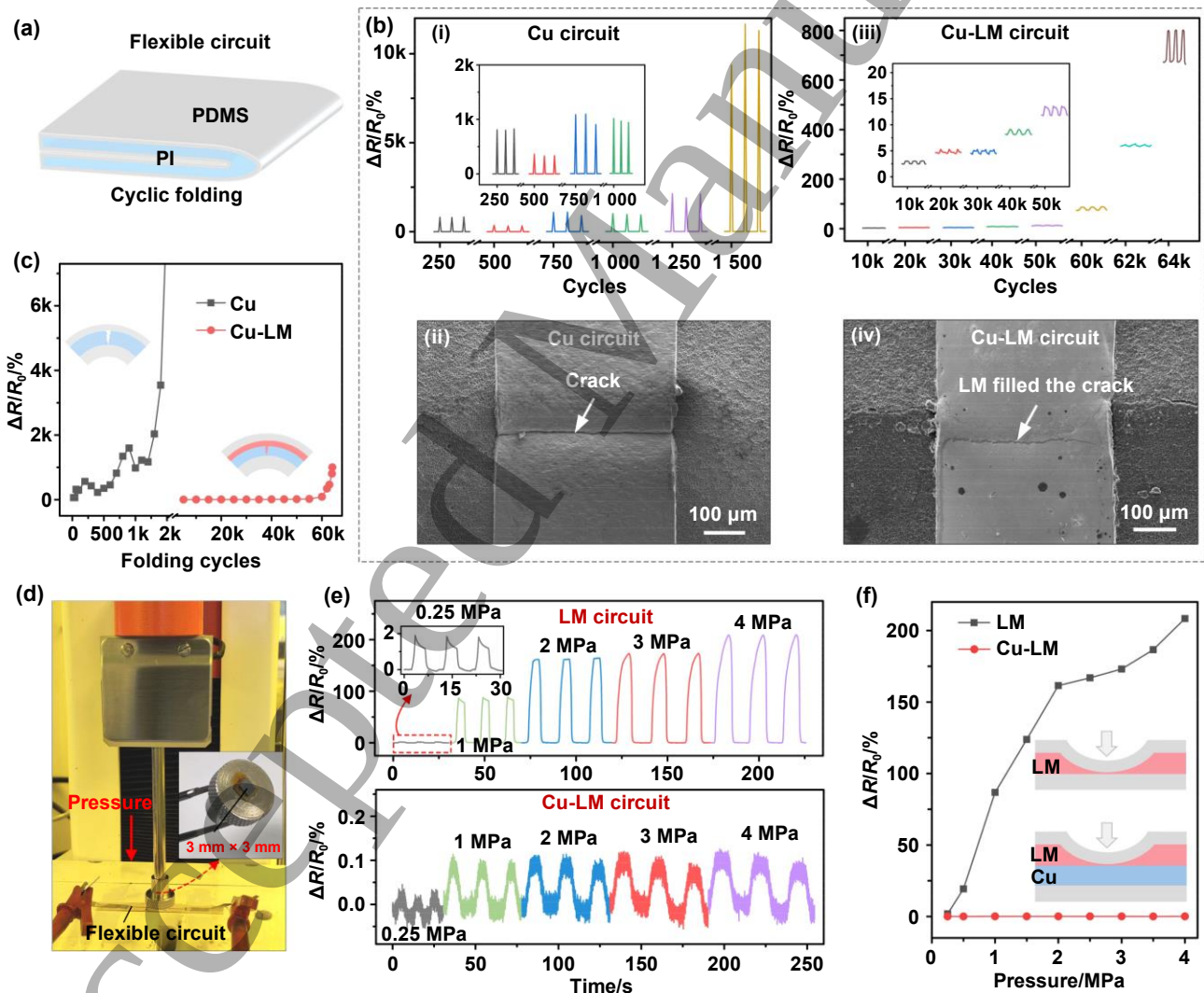


Figure 5. Folding and compression performances of the Cu-LM flexible circuits. (a) Schematic illustration of the cyclic folding test. (b) Resistance changes of the encapsulated circuits and SEM images of the

unencapsulated circuits after the same folding cycles. (i) and (ii) Cu circuit. (iii) and (iv) Cu-LM circuit. (c) Comparison of the long-term folding performances of the Cu circuit and Cu-LM circuit. (d) Image of the compression testing equipment. (e) Resistance changes of the Cu circuit and Cu-LM circuit under different pressures. (f) Comparison of the compression performances of the Cu circuit and Cu-LM circuit.

The real-time healing of cracks ensures improved operational stability and extended lifespan for the flexible circuit, making it particularly suitable for ensuring the stability of long-term continuous signal monitoring of wearable systems that inevitably undergo deformations. It is worth noting that the practical applications of Cu-LM flexible circuits require not only their own stability but also stable interconnections with other electronic components. To this end, we first investigated the connection stability between flexible circuits and electronic components. As a qualitatively intuitive demonstration, the flexible circuit was interconnected with 16 LEDs and standard connectors, forming a flexible 4×4 LED array (Figure 6(a)). This LED array can maintain stable lighting under bending, which preliminarily proves that the flexible circuit can stably connect with other electronic devices. To further quantitatively investigate the connection reliability, we observed the interface status between a resistor (50 Ω) and the flexible circuit and measured the current–voltage (I – V) characteristics under different bending angles. As shown in Figure 6(b), the electrodes at both ends of the resistor can be spontaneously wetted by LM after treatment with HCl vapor [49], as evidenced by the meniscus formed by the wetting of the LM. This demonstrated that flexible circuit could form a good electrical connection with the resistor. This method for stably interconnecting electronic components via LM is also commonly used in other literature [79–81]. The wetting effect occurs because LM undergoes interfacial alloying reactions with the metal coating on the pins of the electronic components. As reported in a previous study [49], for the interface between the Ga-based LM and the pins of rigid electronic components, the possible reactions are mainly the alloying reaction between Ga and gold (Au) or palladium (Pd), while the nickel (Ni) coating can serve as a barrier layer to prevent LM from further reacting with the underlying Cu layer of the pins. Subsequently, the circuit was encapsulated with PDMS, which is intended to ensure the mechanical stability of the electrical connection. To validate the reliability of the interfacial connections, we measured the I – V characteristics under 10°, 60°, 150° bending, and 180° torsion conditions. The resulting four I – V curves overlap well with each other, demonstrating that the flexible circuit can maintain stable interfacial connections with negligible contact resistance variation under varying deformation states.

Taking advantage of the excellent bending resistance and reliable interconnection with other electronic components, this flexible circuit can be further utilized to integrate various electronic components for complex flexible bending-resistant integrated circuits. Here, we constructed a flexible hybrid system for acquiring human pulse signals using flexible circuits (Figure 6(c) and (d)). The corresponding Cu pattern and Cu-LM pattern of the flexible integrated circuits are shown in Figure S17. A resistive pressure sensor for

detecting pulse signals was connected to the sensor interface of the flexible integrated circuit, forming a series circuit with the voltage divider resistor. The voltage across the resistive sensor was sampled by the analog-to-digital converter (ADC) of the microcontroller unit (MCU) and converted into digital signals. Then the data was sent to the host computer via the wireless fidelity (WIFI) module connected to the data output interface of the MCU. Figure 6(e) shows the obtained pulse signals under different states of the flexible circuit. It can be found that the waveform of the collected pulse signals changed negligibly during the bending process, demonstrating the excellent bending stability of the flexible circuit. Furthermore, the flexible integrated circuit was subjected to 10 000 cycles of bending at a radius of 5 mm. After the long-term repeated bending, the pulse signal can still be detected normally. Therefore, the excellent real-time healing capability of the flexible circuit can ensure continuous undisturbed signal detection and prolong the operational lifespan of the flexible integrated system, highlighting its significant potential for applications in long-term wearable health monitoring.

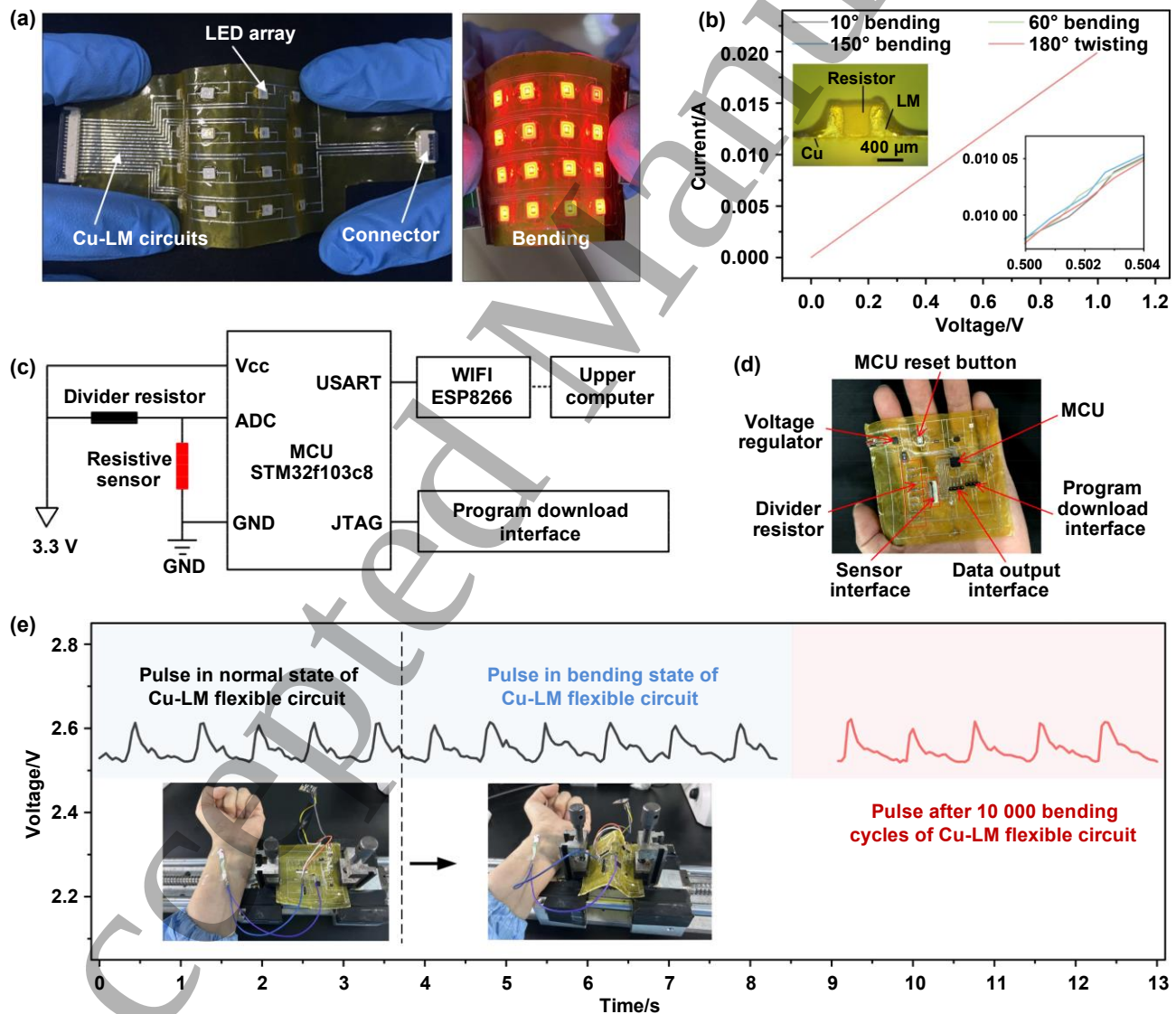


Figure 6. Interconnection reliability and robust flexible integrated system for acquiring human pulse signals.

(a) A 4×4 LED array connected by the flexible circuit. (b) Current–voltage (I – V) characteristics of the circuit connected with a 50 Ω resistor at 10°, 60°, 180° bending, and 180° twisting. (c) Principal diagram of the flexible integrated system for acquiring human pulse signals. (d) Optical image of the flexible system for signal acquisition. (e) Robust bending performance of the flexible acquisition system for human pulse.

3. Conclusion

We have shown an extremely bending-resistant and compression-insensitive flexible circuit achieved by the real-time healing of fatigue cracks in the bottom Cu layer via the spontaneous interfacial wetting of the upper LM. Moreover, the thick bottom Cu layer plays an important role in maintaining conductivity under compression to enhance the compression insensitivity of the circuit. The etched Cu pattern template in the bottom layer also facilitates the printing of LM by selective interfacial wetting, allowing LM to leverage the manufacturing advantages of Cu pattern inherited from conventional photolithography and etching, achieving high-resolution and high-efficiency printing. The obtained flexible circuit can maintain its original stable bilayer morphology over a long time, with alloying reactions occurring only at the interface without further downward penetration. Profiting from the real-time healing of fatigue cracks, compared with the counterpart Cu circuits, the bending-resistant flexible circuit achieved excellent durability of 230 000 cycles under a bending radius of 0.5 mm and 60 000 cycles under repeated folding cycles, achieving improvements of 2.3 and 120 times, respectively, whereas its resistance fluctuation under compression is only 0.05% of that of the counterpart LM circuit. As a demonstration, a flexible signal acquisition system constructed with this circuit can continuously capture pulse signals during bending and remain functional after 10 000 bending cycles. The simultaneously enhanced bending stability and compression insensitivity greatly boost the applicability and reliability of flexible circuits under various deformation conditions.

4. Experimental section

4.1 Materials

Liquid metal (galinstan) with the melting point of 6–9 °C was purchased from Yunnan Zhongxuan Liquid Metal Technology Co., Ltd. Polyimide-copper laminate (PI is 25 μm thick and copper film is 13 μm thick) was purchased from Shenzhen Bangkai Adhesive Products Co., Ltd. PDMS (SYLGARD 184) curing agent and elastomer were purchased from Dow Corning. Sodium hydroxide (NaOH), ferric chloride (FeCl_3), and hydrochloric acid (HCl) were purchased from Aladdin.

4.2 Wet etching Cu patterns

The Cu patterns on PI film were prepared by conventional photolithography and wet etching. The detailed process is as follows. As shown in Figure S3, the commercially available Cu/PI film was cut into the required dimensions and attached to a rigid PET film via electrostatic adsorption to facilitate the

subsequent lithography process. A layer of photoresist (EPG535) was spin-coated on the PI-Cu film (500 rpm for 15 s and then 1 500 rpm for 40 s), followed by baking on a hot plate at 90 °C for 7 min. Then, the sample was exposed to the UV light of a lithography machine for 8 s under the cover of a photomask with customized patterns. Subsequently, it was developed in 0.5 wt% NaOH solution for 12 s. Then, the sample with the photoresist pattern was wet-etched for 40 s at 50 °C in a mixed solution of FeCl₃ (40 wt%) and HCl (37 wt%) with a volume ratio of 3:1, followed by removing the photoresist using acetone and ethanol to obtain the Cu pattern (Figures S2 and S3).

4.3 Fabrication of the Cu-LM flexible circuits

LM was adhered to the Cu patterns by a simple dip-coating method in naturally stratified NaOH (aq) (30 wt%, top layer) and bulk LM (bottom layer). Firstly, LM was dripped into the NaOH solution. During this process, the surface oxides of LM were removed by the NaOH solution, ultimately forming a stratified solution with NaOH solution on the top layer and non-oxidized LM at the bottom layer. Next, dip-coating process was conducted by immersing the Cu pattern into the LM. LM can selectively wet and uniformly coat onto the Cu surface with no residue on the PI surface. Then the flexible circuit was obtained by encapsulating with PDMS (prepolymer and curing agent at a mass ratio of 10:1).

4.4 Characterization and measurements

SEM images were obtained by a field-emission scanning electron microscope (SU8010, HITACHI). The microscopic optical images of the Cu pattern, Cu-LM pattern, and height of LM were obtained by a laser scanning confocal microscope (OLS4000, Olympus). The contact angles were recorded by a contact angle goniometer (Dataphysics, OCA20). To test the long-term bending durability of the Cu-LM circuits and Cu circuits, a customized stretching machine consisting of a stepping motor, a stepping motor controller, a sliding block, a screw, and guide rails was used to perform bending cycles. The bending angle was regulated by controlling the displacements. The cyclic folding tests were conducted by a shaker (SINOCERA JZK-10), which was driven by an excitation signal from the function generator (Agilent 33220A) amplified by a power amplifier (SINOCERA YE5871A). The compression tests were conducted by a uniaxial testing machine (PT1198GTD, POOTAB). The resistance changes and current–voltage (I – V) characteristics during the bending process were monitored by a source meter (B2912A, Keysight).

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Conflict of interest

The authors declare that they have no conflicts of interest.

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